



CodeEmyTM
TECHNOLOGIES

Ignite the Flame of Education

Embedded Systems Engineering Program

Engage in Embedded Systems Mastery

Duration: 3 Months | Mode: Online | Level: Beginner to Advanced
Powered by CodeEmy!



REGISTER NOW



Scope of **Embedded Systems**



15+ million

embedded jobs globally 2026



\$140+ billion

embedded systems market



1.5–2 lakh

job openings in 2026



85%+

manufacturing enterprises

Here is all the **Career Fields** you can excel at

**Embedded Systems
Engineer**

**Embedded Software
Engineer**

Firmware Developer

IoT Embedded Engineer

**Hardware Design
Engineer**

**Automotive Embedded
Engineer**

**Robotics Embedded
Engineer**

**Control Systems
Engineer**

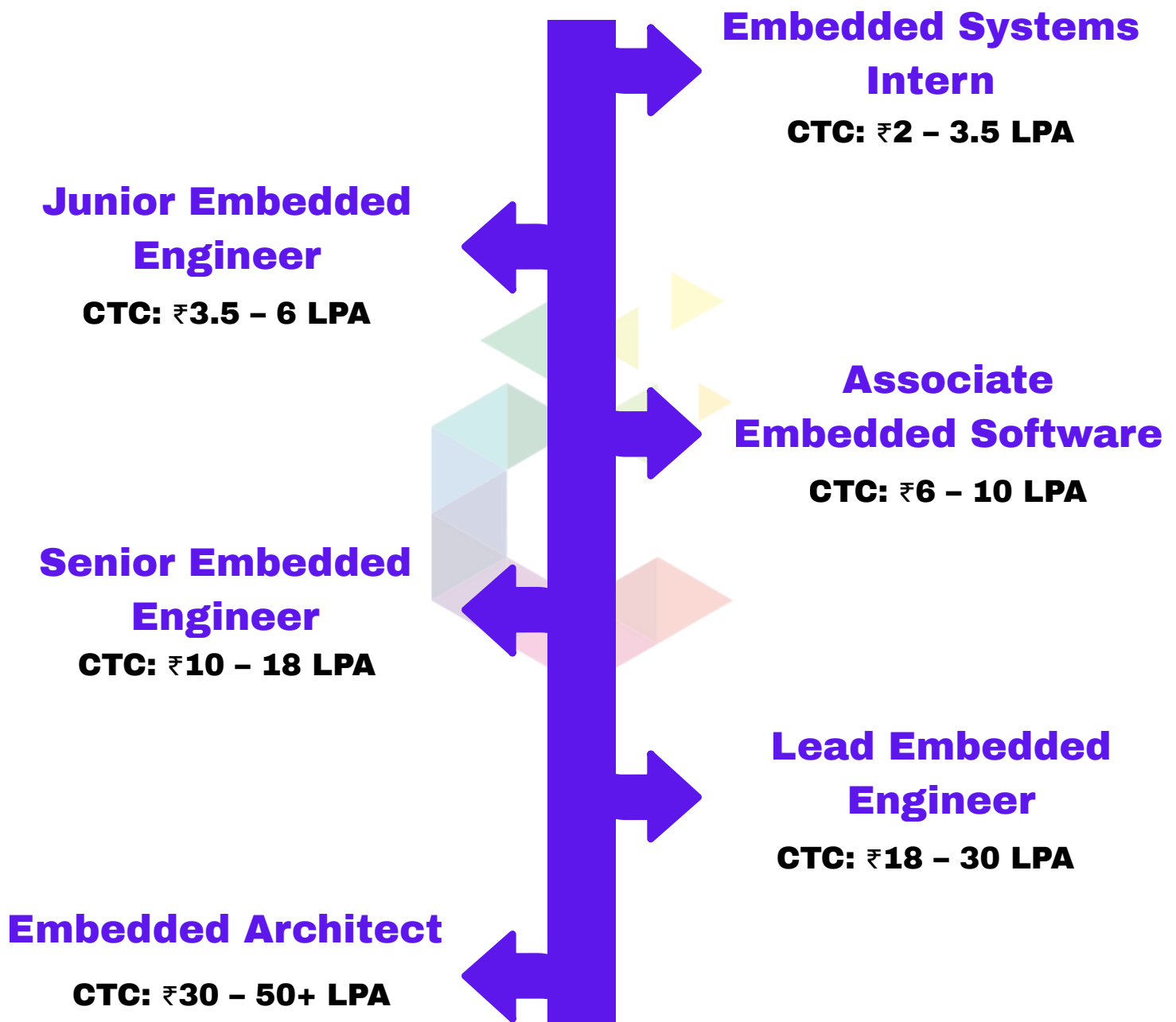
**Product Development
Engineer**

**Embedded Linux
Engineer**

CAREERS



Here is how your **Career Trajectory** will look like



Why **CodeEmy** is Unique ?

Category	CodeEmy	UpGrad	SimpliLearn	Edureka	IIDE
Live Classes	✓	✓	✓	✓	✓
Placement Assistance	✓	✓	✗	✗	✓
Interview Training	✓	✓	✗	✗	✓
LinkedIn Profile & Resume Building	✓	✗	✗	✗	✗
Earn While your Learn	✓	✗	✗	✗	✗
Freelancing	✓	✗	✗	✗	✗
Campus Leader board across the college	✓	✓	✓	✗	✗
Lifetime Access to content	✓	✗	✗	✓	✓
Offline Project Fest (Every 3 Month once)	✓	✗	✗	✗	✗
Course Duration	3 - 4 Months	7-11 Months	5 Months	5 Weeks	4-5 Months
Price	₹15,000	₹ 1,05,000	₹ 1,00,000	₹ 20,000	₹ 72,000

⚠ Disclaimer:

The names of other companies mentioned here are used solely for comparison purposes to highlight the unique features of our offerings. We do not intend to misuse, defame, or misrepresent any organization in any way. There is no affiliation or endorsement implied. Our intent is purely informational and respectful.

UNIQUENESS



Week 1

- What are embedded systems?
- Applications & real-world examples
- Embedded vs general-purpose systems
- Overview of microcontrollers & microprocessors

Project: LED blinking using Arduino

Week 2

- Microcontroller basics (AVR, ARM, PIC)
- CPU, memory, I/O ports, timers
- Interrupts & peripherals

Project: Microcontroller programming basics

Week 3

- Basics of C for embedded systems
- Variables, loops, functions
- GPIO pin control

Project: LED pattern programming

Week 4

- Digital & analog sensors
- Motors, relays, and actuators
- Reading sensor data & controlling actuators

Project: Temperature & motion detection system



Week 5

- UART, SPI, I2C basics
- Wireless communication: Bluetooth, WiFi
- Sensor-to-microcontroller communication

Project: Sensor data transfer via UART/WiFi

Week 6

- Introduction to RTOS (Real-Time Operating System)
- Task scheduling, priorities, and multitasking
- Embedded software lifecycle

Project: Simple multitasking on microcontroller

Week 7

- IoT integration with embedded systems
- Smart devices & automation
- Embedded system design considerations

Project: IoT-based device control (light/fan automation)

Week 8

Major Project:

- Smart Home Automation System
- Embedded Temperature Monitoring System
- Motor Control Robot
- Portfolio development
- Resume & interview guidance



Hands - On Projects

Mini Projects (Beginner to Intermediate Level)

1. LED Pattern Controller

- **Skills Used:** Embedded C, GPIO programming, Arduino
- **Goal:** Create different LED blinking patterns using a microcontroller.

2. Temperature Monitoring System

- **Skills Used:** Sensors (LM35/DHT11), Arduino, LCD interfacing
- **Goal:** Measure temperature and display real-time data on LCD.

3. Automatic Street Light System

- **Skills Used:** LDR sensor, Arduino, Relay control
- **Goal:** Automatically switch street lights ON/OFF based on ambient light.

Major Projects (Advanced Level)

1. Smart Home Automation System

- **Skills Used:** Embedded C/Python, WiFi module, Relays, IoT integration
- **Goal:** Control home appliances remotely using mobile or web interfaces.

2. Industrial Temperature & Gas Monitoring System

- **Skills Used:** Gas sensors, Temperature sensors, Microcontroller, Alerts
- **Goal:** Monitor industrial conditions and trigger alerts during unsafe levels.

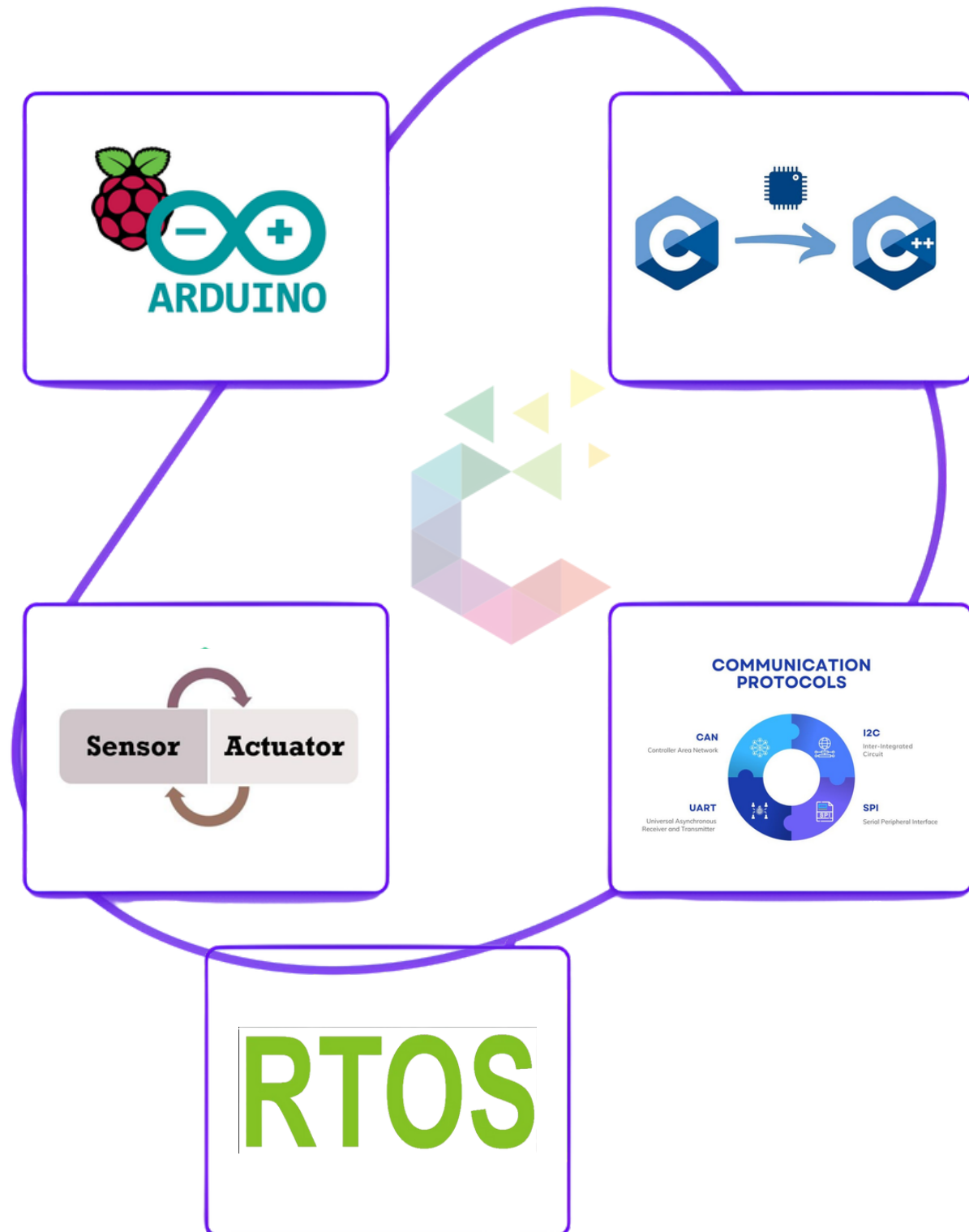
3. Autonomous Robot with Obstacle Avoidance

- **Skills Used:** Ultrasonic sensors, Motor drivers, Embedded programming
- **Goal:** Design a robot that navigates autonomously avoiding obstacles.



Tools

You will learn



Showcase your learning journey

Certificate of Completion from CODEEMY



Certificate of Internship from CODEEMY



Approved by:



Contact Us

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